Chapter 11

An overview of Nduga phonology

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I present an overview of the phonology of Nduga, a Sara-Bongo-Bagirmi language in Central African Republic. Nduga's consonant system includes implosives, prenasalized plosives, prenasalized fricatives, and labial-velar plosives. It has two rhotics, /r ɽ/. Nduga has eight oral monophthongs, one oral diphthong /ᵘa/, and five nasal vowels. Vowel length is contrastive. Oral vowels exhibit vowel harmony between /e o/ and /ɛ ɔ/. Final vowels often reduce to schwa or elide completely. Nduga has three tone levels. All nine combinations of tones occur on bisyllabic nominals. Verb infinitives have lexical tone, with the patterns HH, HM, MM, ML, and LL occurring on bisyllabic verbs.

1 Introduction

In this chapter, I present an overview of the phonology of Nduga (pronounced [ndū.gā]), a speech variety found in the northcentral part of Central African Republic. It is found mostly in the Nana-Grébizi prefecture (Kaga Bandoro subprefecture), but it is also found on the eastern border of Ouham prefecture (Kabo subprefecture). More precisely, it is spoken in and around the villages of Ouandago, Konvi, and Bissekebou (aka Bissikibou) (Moéhama 2021: 7).

Ethnologue considers Nduga to be a dialect of the Luto language (ISO 639-3 code: [ndy]), which as a whole is spoken by about 19,000 people in Central African Republic and Chad (Eberhard et al. 2024). *Ethnologue* lists five dialects of the Luto language: Luto, Nduga, Ndokoa, Wada, and Konga. The first four of these likely correspond to what Boyeldieu et al. (2006) refer to as Luto (lto), Nduga (ndg), Ndoka (ndk), and Wad (wad), respectively. That source provides

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extensive wordlist data for these four varieties.[1] The Nduga data therein were collected by Pierre Nougayrol and include about 460 lexical items.

Nduga is currently classified as part of the Sara-Bongo-Bagirmi (SBB) group within the Central Sudanic family. Dimmendaal et al. (2019: 343–344) consider SBB to be essentially the same as "West Central Sudanic." More general information can be found in *Ethnologue*[2], as well as *Glottolog*[3] (Hammarström et al. 2024).

Some previous research exists on the various varieties of the Luto language. Linguistic descriptions include Moundo (1975) (Luto variety), Ndoko (1991) (Luto variety), and Mololi (2011) (Ndokoa variety). In addition, there is a language survey (Moéhama 2021), demographic information in Nougayrol (1990: 75–77), lexical data in Boyeldieu et al. (2006), a comparative paper looking at the evolution of labial-velar consonants (Olson 2024) (Luto and Nduga varieties), and brief mention in several other articles by Boyeldieu and Nougayrol, as well as Tucker & Bryan (1956: 17) (Ndokoa dialect) and Bender (1997: 34). To my knowledge, the current work is the first to focus on Nduga phonology.

My research was conducted in Bangui, Central African Republic, during three visits from 2012 to 2017, for a total of about four weeks. I worked with a team of three speakers, who are fluent in both the Luto and Nduga dialects. Unless otherwise noted, the data in the paper are from our consulting sessions. Part of the research was done in collaboration with Paul & Jo Murrell and Brad & Maria Festen.

We collected mostly lexical data, but also some paradigms and texts. The wordlist in the appendix is based on the 204-item wordlist tool in Moñino (1988).

I discuss consonants in Section 2, vowels in Section 3, tone in Section 4, syllable patterns in Section 5, and word patterns in Section 6. Additional discussion is provided in Section 7, including addressing the question of how Nduga relates to the rest of SBB. Concluding remarks are given in Section 8.

Phonemic forms are enclosed by forward slashes, e.g. /ᵑg͡b/, and phonetic forms are enclosed by square brackets, e.g. [ŋgb]. Superscripts and tie bars are used in phonemic forms in order to help distinguish them visually from phonetic forms, as well as to help show the unitary nature of the phonemic digraphs and trigraphs. The phonetic data are transcribed using the extant International Phonetic Alphabet (IPA 2006).

[1] The data from Boyeldieu et al. (2006) can be downloaded from the following URL: https://en.wiktionary.org/wiki/Appendix:Proto-Sara-Bongo-Bagirmi_reconstructions.

[2] https://www.ethnologue.com/language/ndy

[3] https://glottolog.org/resource/languoid/id/luto1241

2 Consonants

Nduga has 28 consonant phonemes, as shown in Table 1.

Table 1: Consonant phonemes in Nduga

	Labial	Alveolar	Retr./Pal.	Velar	Labial-velar	Glottal
Implosive	ɓ	ɗ				
Plosive	(p)	t		k	(k͡p)	
	(b)	d		g	g͡b	
	(ᵐb)	ⁿd		ᵑg	(ᵑg͡b)	
Fricative	(f)	s				(h)
	v	z				
	ᵐv	ⁿz				
Nasal	m	n				
Trill/flap		r	ɽ			
Lateral		l				
Approximant			j		w	

Labial-velar and prenasalized consonants are written with two or three symbols, but they pattern as unitary phonemes in Nduga. Parentheses indicate phonemes that are marginal to the phonological system.

Sample contrasts between the consonants within the same broad category of place of articulation are given in Table 2. Insofar as possible, I provide contrast between nouns in word-initial position with a following /a/.

The period (full stop) symbol < . > indicates a syllable break. The hyphen < - > indicates a morpheme break. Glosses were obtained in French and translated into English for this paper. Corresponding entries in the wordlist are indicated by the number sign < # > followed by the entry number. Corresponding entries in the Boyeldieu et al. (2006) wordlist are given as N/[number] for nouns and V/[number] for verbs. Transcriptions from that source are unmodified.

Several phonemes are rare: /p, ᵐb, h/ do not occur in the wordlist, while /b, f, k͡p, ᵑg͡b/ occur only once or twice. These phonemes are also the rarest ones in Nougayrol's data, with 10 or fewer occurrences of each phoneme. The implosives /ɓ, ɗ/ are more common than their plain counterparts /b, d/ in the wordlist.

Boyeldieu et al. (2006) include a retroflex lateral [ɭ] in their transcriptions of the Luto varieties. I initially employed the same transcription. However, examination of the acoustic properties (cf. Figure 1) showed that the sound exhibits a short

Table 2: Consonant contrasts in Nduga

	Phoneme	Phone	Example	Gloss	No.
Labial	/ɓ/	[ɓ]	[ɓā.rà]	rainy season	# 163
	/p/	[p]	[pì.píī]	hot pepper	
	/b/	[b]	[bā.ŋgàm]	sweet potato	
	/^{m}b/	[mb]	[mbà.mbò]	one hundred	
	/f/	[f]	[fã́ã̄]	type of tuber	
	/v/	[v]	[và.ɗù]	pig	
	/^{m}v/	[ɱv]	[ɱvá.lè]	flute (from horn)	
	/m/	[m]	[mà.nɛ̀]	water	# 54
	/w/	[w]	[wā]	granary	
Alveolar	/ɗ/	[ɗ]	[ɗá.rə́ tà.rú]	to stand	# 47
	/t/	[t]	[tā.rà]	mouth	# 14
	/d/	[d]	[dā.lā]	drum	
	/^{n}d/	[nd]	[ndà.rā]	skin	# 131
	/s/	[s]	[sà.ná]	type of basket	
	/z/	[z]	[zā.rù]	civet	
	/^{n}z/	[nz]	[nzā.mè]	ground squirrel	
	/n/	[n]	[nā.nā] ~ [nā.nū]	maternal uncle	# 126
	/l/	[l]	[là.mú]	clothing	
	/r/	[r]	[rā]	sky	
Retr./Pal.	/ɽ/	[ɽ]	[ɽā.ɗà]	type of net	
	/j/	[j]	[jà.kú]	quick, fast	
Velar	/k/	[k]	[kā.gā]	tree	# 9
	/g/	[g]	[gā.zù]	horn	# 36
	/^{ŋ}g/	[ŋg]	[ŋgá.ɽā]	ground	# 178
Lab.-velar	/k͡p/	[kp]	[kpā.rù]	poison	
	/g͡b/	[gb]	[gbā.gū]	wing	# 3
	/^{ŋ}g͡b/	[ŋgb]	[ŋgbā.ɽā]	assegai	# 162
Glottal	/h/	[h]	[hà]	pour	

closure period of about 30 ms, which is indicative of a tap or flap (Catford 1977: 130) rather than a lateral. In addition, there is a significant reduction in intensity during closure. These properties lead me to consider the sound to be a retroflex flap [ɽ] and transcribe it accordingly.

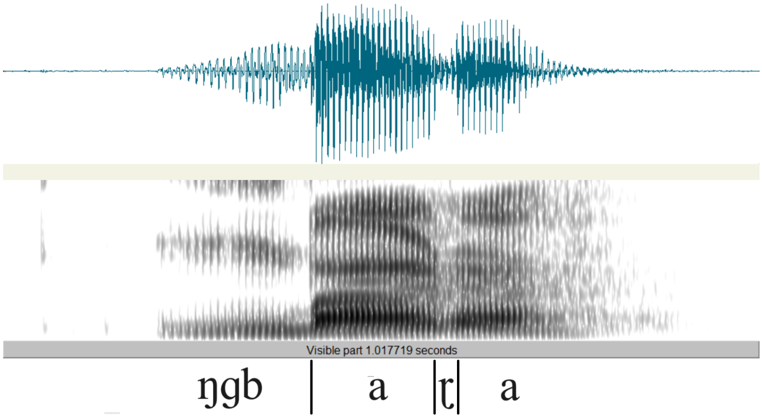

Figure 1: Waveform and spectrogram of the word [ŋgbā.ɽā] ‘assegai’ (# 162)

In word-initial position, the retroflex flap [ɽ] is preceded by a short vowel-like segment, as shown in Figure 2. This prothetic segment is discussed by Ladefoged & Maddieson (1996: 240–241) for a post-alveolar flap in Walpiri. They state that the “segment assists in setting up the acoustic cues for the flap, which thus become comparable in both initial and intervocalic environments” (p. 240).

A similar prothetic segment has been observed for the word-initial labiodental flap [ⱱ] in Mono (Olson 2005: 126). It has also been attested for alveolar trills in several languages (Bolter 2021) (cf. Ladefoged & Maddieson 1996: 219 for Italian). However, the alveolar trill /r/ in Nduga is not accompanied by this prothetic segment.

The phoneme /j/ is nasalized before a nasal vowel, as shown in (1):

(1) Nasalization of /j/
/j/ → [j̃] / _Ṽ

The nasalized allophone [j̃] could potentially be interpreted instead as a phoneme /ɲ/ which causes a following vowel to become nasalized. Boyeldieu et al. (2006) implicitly follow this analysis as they employ the symbol [ɲ] for the sound in their transcriptions of the Luto varieties.

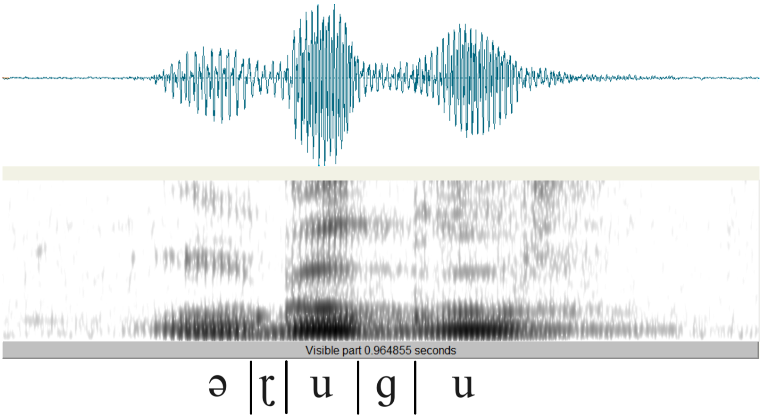

Figure 2: Waveform and spectrogram of the word [ɽū.gū] 'to burn (transitive)' (# 18), with prothetic vowel-like segment

However, I opt to interpret [j̃] as an allophone of /j/ for a several reasons. First, nasal vowel phonemes are independently motivated in Nduga (cf. Section 3). Second, I did not find other cases of nasalization on vowels occurring adjacent to a nasal consonant. Third, the speakers' intuitions were that [j] and [j̃] comprise a single phoneme. Fourth, the Nduga speakers noted that the tongue does not touch the roof of the mouth during the articulation of the sound. Phonetically, the sound is [j̃], not [ɲ].

Sample words that show the nasalization of /j/ are given in Table 3:

Table 3: Sample words with nasalized [j̃]

Phonemic	Phonetic	Gloss	Number
/vī.jã̄/	[vī.j̃ã̄]	goat	# 27
/zɔ̄.jɔ̃̀/	[zɔ̄.j̃ɔ̃̀]	to bite	# 110
/jɛ̃́.lɛ̃̀/	[j̃ɛ̃́.lɛ̃̀]	wind	# 195
/jã̀.lú/	[j̃ã̀.lú]	rheum	
/jã̀.ɓū/	[j̃ã̀.ɓū]	shea fruit	

There is possibly a comparable rule for /w/, but our data are limited to one example: /kũ̄.wũ̀/ [kũ̄.w̃ũ̀] 'to drink' (# 12).

3 Vowels

Nduga has nine oral vowels and five nasal vowels. The oral vowels include eight monophthongs and one diphthong, as shown in Table 4.

Table 4: Vowel phonemes in Nduga

	Front	Central	Back
Oral	i		u
	e	ə	o
	ɛ		ɔ
		a	(ᵘa)
Nasal	(ĩ)		(ũ)
	(ɛ̃)		(ɔ̃)
		(ã)	

Contrast between the oral vowels is given in Table 5. I provide contrast between vowels immediately following a word-initial /k/:

Table 5: Oral vowel contrasts in Nduga

Phoneme	Phone	Transcr.	Gloss	Number
/i/	[i]	[k-ī.lì]	to be black	# 117
/e/	[e]	[kē.mè]	stomach	# 196
/ɛ/	[ɛ]	[k-ɛ́.ɽɛ̄]	to collect water, to bail	
/ə/	[ə]	[kə̀]	(grammatical particle)	
/a/	[a]	[kā.là]	alone, separately	
/u/	[u]	[kú.lù]	charcoal	# 24
/o/	[o]	[kō.lō]	type of tree	
/ɔ/	[ɔ]	[kɔ̄.lɔ̀]	giraffe	
/ᵘa/	[u̯a]	[k-u̯à]	to die	# 113

Nduga exhibits vowel harmony: The vowels /e o/ do not cooccur with /ɛ ɔ/ in the same root. There is very little affixation in Nduga, so the root is usually the same as the word.

Table 6: Cases of /ə/ and /^{u}a/ in the wordlist

Phoneme	Transcription	Gloss	Number
/ə/	[ɗá.rə́ tà.rú]	to stand (stay + upright)	# 47
	[mə̄]	1SG.SBJ	# 96
	[ŋgé.ŋgə̄ sá.kū]	panther (lion + little)	# 130
/^{u}a/	[k-ū.gu̯à]	to cut (with an axe)	# 39
	[nzū.ŋgu̯à]	twin	# 97
	[k-u̯à]	to die	# 113
	[ní.ŋgá ku̯â]	snake (thing + die)	# 172

The vowels /ə/ and /^{u}a/ are somewhat rare in the wordlist, only occurring in a handful of words, as shown in Table 6.

The rarity of /ə/ in the wordlist is due to its absence in monomorphemic words in major grammatical categories. Most of its occurrences involve the reduction of another vowel to schwa in word-final position before other words in a noun or verb phrase, as shown in (2).

(2) Vowel Reduction (before another word within a phrase)
V → ə / _# word

Examples of Vowel Reduction are shown in Table 7.

Table 7: Examples of vowel reduction

UR	SR	No.
[kā.gā] 'tree' + [gē] 'PL'	⟶ [kā.gə̄ gē] 'trees'	
[bē.gī] 'shoulder' + [má] '1SG.POSS'	⟶ [bē.gə̄ má] 'my shoulder'	
[ŋgé.ŋgē] 'lion' + [sá.kū] 'little'	⟶ [ŋgé.ŋgə̄ sá.kū] 'panther'	# 130

In some cases, the reduced vowel is instead completely elided, as shown in Table 8. Mololi (2011: 106) describes a similar process involving the plural morpheme in the Ndokoa variety.

The only cases where schwa is not a reduced form of another vowel involve grammatical function words. Examples are shown in Table 9.

This situation, in which a phoneme has a robust presence only in grammatical particles, is attested elsewhere. Parker (2009) mentions three languages in

Table 8: Example of Vowel Elision

UR		SR	No.
/ɗá.rá/ ‘remain’ (V/094) + /tà.rú/ ‘standing’ (N/063)	→	[ɗá.rə́ tà.rú] ‘to stand’ ~ [ɗár tà.rú]	# 47

Table 9: Schwa in grammatical function words

Phonetic	Gloss	Number
[mə̄]	1SG.SBJ	# 96
[nzə̀]	PFV	
[má.nə̄]	FUT	
[sə̀]	this	
[kə̀]	(grammatical particle)	

which a consonant phoneme only occurs in an affix, and never in a lexical root: Arammba [stk] (PNG), Awara [awx] (PNG), and Guambiano [gum] (Colombia), cf. Quigley (2003), Branks & Branks (1973).

I consider /ᵘa/ [u̯a] to be a diphthong for several reasons. First, there are no other syllable medial approximant-vowel phonetic sequences in Nduga. Second, the approximant portion [u̯] of the phoneme is phonetically short. Third, on the two words in the wordlist that have falling tones on the phoneme ([k-u̯à] ‘to die’ # 113, [ní.ŋgá ku̯â] ‘snake’ # 172), the tone occurs on the [a] portion of the phoneme.

Figure 3 shows an F1 vs. F2 plot of the Nduga oral monophthongs. Ten tokens of each vowel are plotted. A vowel symbol is given for each individual token. The axes are marked in Hertz but scaled on the Bark scale. The ellipses are centered on the mean for each vowel and have radii of two standard deviations (Maddieson & Anderson 1994). There is reasonable separation between most of the vowels, indicating that the values of F1 and F2 are sufficient acoustic properties for distinguishing the vowels.

In addition to the oral vowels, Nduga also has a series of nasal vowels [Ṽ] and long vowels [V:]. Not enough data were collected to provide contrast in the same environment for all possible nasal and long vowels. Tables 10 and 11 provide samples of the sounds.

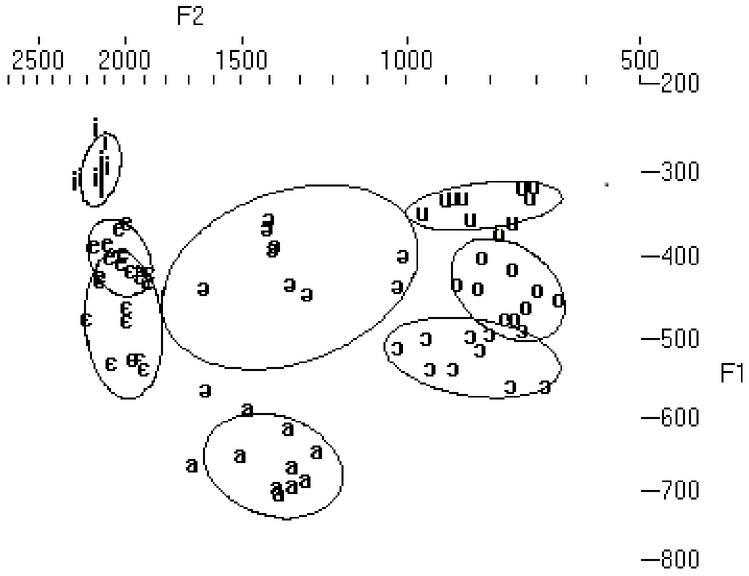

Figure 3: Plot of F1 vs. F2 for Nduga oral vowels

Table 10: Nasal vowels in Nduga

Phoneme	Transcription	Gloss	Number
/ĩ/	[j̃ĩ̄ĩ̀]	scorpion	
/ɛ̃/	[j̃ɛ̃́.lɛ̀]	wind	# 195
/ã/	[zã̄]	animal	# 6
	[hã̀]	galago	
	[fã́ã̄]	type of tuber	
/ũ/	[k-ũ̄.w̃ũ̀]	to drink	# 12
/ɔ̃/	[zɔ̄.j̃ɔ̃̀]	to bite	# 110
	[nɔ̃̄ɔ̃̄]	tears	# 141
	[nɔ̃́]	other	

Table 11: Long vowels in Nduga

Phoneme	Transcription	Gloss	Number
/iː/	[mìí]	five	# 30
/eː/	—		
/ɛː/	—		
/aː/	[nzāà]	white	# 11
	[sāà]	smoke	# 82
	[māā]	thigh	# 43
	[bāà]	papa	# 132
	[kàà]	mother-in-law	
	[ɱvāà]	tsetse fly	
/uː/	[k-ūù]	to swell	# 59
/oː/	[kóò]	mother	# 108
	[tōō]	to blow	# 174
	[kóó tì.gì]	bee (mother + honey)	# 1
/ɔː/	[ɗɔ̄ɔ̄]	war	# 89
	[sɔ̀ɔ́]	four	# 151
	[sɔ̄ɔ̄]	type of trap	

Two lexical items in our data contain vowels which are both nasalized and lengthened, shown in Table 12.

Table 12: Long nasal vowels in Nduga

Phoneme	Transcription	Gloss	Number
/ãː/	[fã́ã̄]	type of tuber	
/ɔ̃ː/	[nɔ̃̄ɔ̃̄]	tears	# 141

The long vowels could be interpreted as either phonemic long vowels (Vː) or sequences of identical vowels (V_1V_1). I have chosen the first option, because there are no cases of dissimilar vowel sequences (i.e. V_1V_2) in Nduga, given that [u̯a] is considered to be a diphthong.

The long vowels often bear two distinct level tones (cf. Table 11). The tone patterns attested on long vowels are: HH HM HL MM ML LH LL. Since long vowels can bear two tones, I generally write them with two identical vowel symbols in order to allow marking the two tones in a clear manner.

Short vowels sometimes have contour tones, which I consider to be sequences of level tones. This is discussed further in Section 4.

4 Tone

Nduga has three robust level tones: high (H), mid (M), and low (L). Sample contrasts are shown in Table 13.

Table 13: Tone contrasts in Nduga

	Transcription	Gloss	Number
H vs. M	[kpɔ́.rɔ̀]	flavor of a salt substitute	
	[kpɔ̄.rɔ̀]	path	
H vs. L	[kī.lá]	tail	# 152
	[kū.là]	work	# 187
M vs. L	[kī.rī]	bundle of sticks	# 67
	[kē.rì]	type of bird	

In addition, several contour tones occur on short vowels. If these are analyzed as sequences of level tones, the patterns HM HL and ML are attested, as shown in Table 14.

Table 14: Contour tones in Nduga

	Transcription	Gloss	Number
HM	[já᷇]	new	# 115
HL	[wâ]	sand	# 161
	[kí.ɽí.kî]	knee	# 84
	[ní.ŋgá ku̯â]	snake	# 172
ML	[k-u̯a᷆]	to die	# 113
	[na᷆]	who	# 153
	[k-ū.lì ɱva᷆]	to nurse	# 180
	[ɓi᷆]	to fly	# 200

In the subsections that follow, I present several observations about tone in Nduga. There are likely other tonal phenomena in the language, which I leave for future research.

4.1 Lexical tone on nouns

Nouns have lexical tone. Eight of the nine possible combinations of tones occur on bisyllabic nouns, as shown in Table 15. The additional pattern (HH) is not attested on nouns, but it does occur on the numeral [zí.jó] ‘two’ (# 49).

Table 15: Tone patterns on bisyllabic nominals

	Transcription	Gloss	Number
HH	[zí.jó]	two	# 49
HM	[kú.nū]	nose	# 116
HL	[kú.lù]	charcoal	# 24
MH	[kū.nzá]	chicken	# 146
MM	[ŋgā.ŋgā]	tooth	# 48
ML	[kā.ɓè]	egg	# 124
LH	[mà.sú]	blood	# 166
LM	[nì.jē]	female	# 70
LL	[mà.nɛ̀]	water	# 54

4.2 Lexical tone on verbs

Verbs exhibit lexical tone on infinitives. Five of the nine possible combinations of tones occur on bisyllabic verb infinitives, as shown in Table 16.

Table 16: Tone patterns on bisyllabic verb infinitives

	Transcription	Gloss	Number
HH	[tá.zá]	to count	# 33
HM	[nzí.lē]	to say	# 50
HL	—		
MH	—		
MM	[ndī.ɗī]	to cook	# 42
ML	[tɔ̄.ɗɔ̀]	to sleep	# 52
LH	—		
LM	—		
LL	[gè.ɽì]	to know	# 34

The most common patterns on infinitives are ML and LL. We did not find any infinitives that exhibit a pattern in which the second tone is higher than the first.

4.3 Tone on pronouns

Subject and object pronouns bear M tones on all syllables. For the corresponding possessive pronouns, the first syllable is H instead of M, as shown in Table 17.

Table 17: Tone patterns on pronouns

Person	Subject	Object	Inalienable Possessive	Alienable Possessive
1SG	mə̄	mā	NOUN + má	NOUN + nā má
2SG	jī	jī	NOUN + i	NOUN + nā jí
3SG	rō	rō	NOUN + ró	NOUN + nā ró
1PL.INCL	zī.gī	zī.gī	NOUN + zí.gī	NOUN + nā zí.gī
1PL.EXCL	zē	zē	NOUN + zé	NOUN + nā zé
2PL	sē.gē	sē.gē	NOUN + sé.gē	NOUN + nā sé.gē
3PL	rō.gē	rō.gē	NOUN + ró.gē	NOUN + nā ró.gē

For the 2SG inalienable possessive pronoun, the vowel /i/ replaces the final vowel of the noun and takes its tone, e.g. [kú.nū] 'nose' + [i] '2SG.POSS' ⟶ [kú.nī] 'your (sg.) nose'.

4.4 H-spread

For nouns with a /HL/ tone pattern, the L changes to H when the plural clitic /gē/ is attached, as shown in Table 18.

Table 18: Examples of H-spread in Nduga

Singular	Plural	Gloss	Number
[mú.mù]	[mú.mú gē]	fog	# 16
[kú.lù]	[kú.lú gē]	charcoal	# 24
[ɗɔ́.vɔ̀]	[ɗɔ́.vɔ́ gē]	road	# 26
[bí.sì]	[bí.sí gē]	dog	# 28
[ká.ɱvà]	[ká.ɱvá gē]	leaf	# 73

In an autosegmental analysis, this would likely be construed as the spread of the H from the first syllable to the second syllable, accompanied by the delinking of the L.

More research is necessary to determine if this process is more widespread in Nduga.

5 Syllable patterns

Nduga has the following syllable patterns: V, CV, CṼ, CVː, CṼː, Cu̯a, and CVC, as shown in Table 19. The most common syllable pattern is CV.

Table 19: Nduga syllable patterns

	Transcription	Gloss	Number
V	[á.nɛ̄]	this	# 19
CV	[zī]	arm	# 15
CṼ	[zã̄]	animal	# 6
CVː	[mìí]	five	# 30
CṼː	[fã́ã̄]	type of tuber	
Cu̯a	[k-u̯à]	to die	# 113
	[nzū.ŋgu̯à]	twin	# 97
CVC	[kèm]	in	
	[kàm]	in front of	
	[bà.ŋgàm]	sweet potato	N/757
	[gbèr.ŋgē]	support (put on head to carry wood)	

Since Cu̯a includes a diphthong (as discussed in Section 3), it could be considered to fit in the CV syllable pattern. The syllable patterns Cu̯a and CVC do not occur with nasal or long vowels.

Most cases of the CVC syllable pattern result from the loss of a word-final vowel in a CV.CV word pattern, as shown in Table 20.

The coda position in the CVC syllable pattern is usually filled by a sonorant. In one case in the wordlist, a fricative occurs in the coda position due to the elision of a word-final vowel: [k-ō.sò] 'pierce' (# 186) + [ká] 'action of crying' ⟶ [k-ōs.-ká] 'to cry' (# 141).

Consonant clusters are very rare in Nduga, and they always span a syllable boundary. The only example I found that may be monomorphemic was [gbèr.ŋgē] 'support'.

Table 20: Sample origins of words with CVC syllable pattern

Transcription	Gloss	Source
[kèm]	in	< [kè.mè] ‘stomach’ (# 196)
[kàm]	in front of, spread out on	< [kā.mù] ‘eye’ (# 123)
[ɓól̄] ~ [ɓó.lō]	to be bitter (# 5)	< SBB *ɓɔlɔ (N/370)
[ɓāl] ~ [ɓā.là]	year (# 7)	< SBB *ɓala (N/355)

An alternative analysis that might fit the data would involve treating the CV: and CṼ: patterns as bisyllabic. In this case, CV: would be divided into CV and V, and CṼ: would be divided into CṼ and Ṽ. This would reduce the number of syllable patterns in the language by one while maintaining the same number of word patterns. One advantage of my analysis is that it matches native speaker intuitions.

A couple of additional phonetic syllable patterns occur very rarely: [Ci̯a] and [Cu̯i]. These occur as the result of the shortening of some two-syllable words to one syllable in casual speech, as shown in Table 21.

Table 21: Examples of [Ci̯a] and [Cu̯i] syllables

		Gloss
[ŋgī.jà mā]	→ [ŋgi̯à mā]	myself
[ɽī.jā]	→ [ɽi̯ā]	type of gourd
[gū.jī]	→ [gu̯ī]	beyond

6 Word patterns

Nduga has the word patterns shown in Table 22. The most common word pattern is CV.CV.

There is a robust set of monosyllabic lexical words in Nduga. Examples are shown in Table 23. This indicates that there is no minimal word restriction in the language.

The V syllable pattern only occurs word-initially. The CV: and Cu̯a syllable patterns only occur word-finally. Many of the words that we originally thought had more than two syllables turned out to be multimorphemic. For example,

Table 22: Word patterns in Nduga

	Transcription	Gloss	No.
CV	[zī]	arm/hand	# 15
CṼ	[zā̃]	animal	# 6
CVː	[mìí]	five	# 30
CṼː	[nɔ̃̄ɔ̃̄]	tears	# 141
Cu̯a	[k-u̯â]	to die	# 113
CVC	[kèm]	in	
V.CV	[ā.nɛ̀]	this	# 19
V.Cu̯a	[ù.ku̯à]	take (cf. V/088)	# 80
CV.CV	[kā.gā]	tree	# 9
CṼ.CV	[ɟɛ̃́.lɛ̀]	wind	# 195
CV.CṼ	[vī.ɟā̃]	goat	# 27
CV.CVː	[pì.píī]	hot pepper	
CV.Cu̯a	[k-ū.gu̯à]	to chop (axe)	# 39
CVC.CV	[gbèr.ŋgē]	support (cloth put on head to carry wood)	
CV.CVC	[bā.ŋgàm]	sweet potato	
CV.CV.CV	[kí.ndà.gú]	star	# 63

Table 23: Monosyllabic lexical words in Nduga

Transcription	Gloss	No.
[zā̃]	animal	# 6
[zī]	arm	# 15
[tá]	to finish	# 75
[wà]	heavy	# 102
[ɽī]	name	# 118
[dè]	person	# 133
[wâ]	sand	# 161
[zò]	head	# 179
[ɓē]	village	# 198

[nzà.kú.zì] ‘bottom’ (# 44) likely derives from [nzà] ‘foot’ (# 136) + [kú.zì] ‘house’ (# 104). Four 3-syllable words in the wordlist appear to be monomorphemic: [kí.ndà.gú] ‘star’ (# 63), [kí.ɽí.kî] ‘knee’ (# 84), [dà.vú.là] ‘man’ (# 91), and [ɽí.kí.zī] ‘nail/claw’ (# 127).

As mentioned in Section 5, most words with a CVC word pattern can be traced to words with a CV.CV word pattern. For example, the function words [kèm] ‘in’ and [kàm] ‘in front of’ appear to come from [kē.mè] ‘stomach’ (# 196) and [kā.mù] ‘eye’ (# 123), respectively. Lexical words with an optional CVC pattern are more commonly pronounced CV.CV, e.g. [ɓól̄] ~ [ɓó.lō] ‘be bitter (# 5) and [ɓāl] ~ [ɓā.là] ‘year’ (# 7). The Proto-SBB pattern for these words was CV.CV: *ɓɔlɔ (N/370) and *ɓala (N/355), respectively.

7 Discussion

Nduga is well-situated within the SBB family (Boyeldieu 2006; Dimmendaal et al. 2019). It has prenasalized stops, labial-velar stops, and implosives, all of which are common in SBB. It has closed syllables that result from the erosion of a final vowel *CVCV > CVC. One unique aspect of its consonant inventory is the absence of /tʃ/ and /dʒ/.

Nduga shares some characteristics with Sara and others with Bongo-Bagirmi. Like Sara, it has contrastive nasal vowels and a three-tone system. Like Bongo-Bagirmi, on the other hand, it has vowel harmony and a more robust set of fricatives than is typical of Sara. This harmonizes with Boyeldieu (2006: 4)’s view that Nduga is closely related to Sara, but it forms a separate branch.

Most verbs in our data begin with [k]. The same verbs in the wordlist found in Boyeldieu et al. (2006) are given as vowel initial, without the [k]. Some forms from the two sources are compared in Table 24.

This suggests that when the root has an initial vowel, the infinitive is marked by a prefix /k-/, and when the root has an initial consonant it is marked as /∅-/.

The very last example in Table 24 shows a case where the initial /k/ appears to be an integral part of the verb root and not a prefix.

8 Conclusion

This paper makes a significant contribution to our understanding of a speech variety that has received very little attention in the literature. I’ve provided information about Nduga consonants, vowels, tone, syllable patterns, and word patterns.

Table 24: Nduga words that begin with /k-/ in our wordlist, with corresponding forms in Boyeldieu et al. (2006)

Our data	Boyeldieu et al.	Gloss	Number	Boyeldieu ref. no.
k-à.ɓà	àɓ	to go	# 4	V/051
k-ù.zà	ùzà (parfois ūzà)	to attach	# 10	V/011
k-ū.jà	ūjà	to cut	# 38	V/098
k-ū.ɗì	ūɗì	to dig	# 41	V/241
k-à.dà	àd	to give	# 51	V/024
k-ō.jò	ɔ̄y`	to give birth	# 58	V/147
k-ɔ̀	ò (ò: ?)	to hear	# 60	V/018
k-ū.là	ūlà	to send	# 61	V/085
k-ì.nà	ìnà	to throw	# 98	V/203
ké.sī	ké.sū	to cough	# 184	V/110

At the same time, it is a modest first step in the large task of describing a language. There are many possible topics for further research. The tonal system of Nduga warrants a more in-depth investigation, especially how tone is realized in the tense-aspect-mood system. A deeper examination of H-spreading could determine if the process extends to other environments. Examining in more detail the relationship between the extant Nduga forms and those of Proto-SBB would inform our understanding of the historical development of the language. The occurrence of vowel reduction/elision at the phrasal level suggests the possibility of metrical structure in the language. The H-spreading process suggests that tonal feet may be present. Finally, it would be fruitful to examine morpho-syntactically complex forms in more detail to uncover possible additional phonological alternations in the language.

Abbreviations

Abbreviations in this chapter follow the Leipzig Glossing Rules, with the following additions.

H	High tone	n.	Noun
L	Low tone	No.	Number
M	Mid tone	Pal.	Palatal
C	Consonant	PNG	Papua New Guinea
Lab.	Labial	ref.	Reference

Retr.	Retroflex	UR	Underlying representation
SBB	Sara, Bongo, Bagirmi	V	Vowel
SR	Surface representation	vb.	Verb

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Appendix A Nduga Wordlist

In this appendix, I present a 204-item wordlist of Nduga (ISO 639-3 code [ndy]). It is based on the 204-item wordlist tool from Moñino (1988). The French glosses are from the source and were used as prompts in collecting the data. The English translations are mine.

Table 25: 204-item wordlist (001–034)

	French	English	Phonetic form	Notes
001	abeille	bee	kóó tì.gì	‘mother’ + ‘honey’
002	acide (vb.)	to be tart	ɗú.sū	
003	aile	wing	gbā.gū	
004	aller	to go	k-à.ɓà	
005	amer (vb.)	to be bitter	ɓó.lō	
006	animal	animal	zã̄	
007	année	year	ɓā.là	
008	appeler	to call	ɓɔ̄.rɔ̄	
009	arbre	tree	kā.gā	
010	attacher; lier	to attach	k-ù.zà	
011	blanc	to be white	nzāà	
012	boire	to drink	k-ũ̄.w̃ũ̀	
013	bon (être)	to be good	gà.ɽà	
014	bouche	mouth	tā.rà	
015	bras	arm	zī	
016	brouillard	mist; fog	mú.mù	
017	brûler (INTR)	to burn (INTR)	ɽū.gū	
018	brûler (TR)	to burn (TR)	ɽū.gū	
019	ceci	this	á.nɛ̄	
020	cela	that	á.nɛ̄ ɗá	
021	ce (en question)	this	á.nɛ̄	
022	cendres	ash	vī.tī	
023	champ	garden	mē.gè	
024	charbon	charcoal	kú.lù	
025	chaud	to be hot	tù.mà ~ k-ù.mà	
026	chemin	path	ɗɔ́.vɔ̀	
027	chèvre / bouc	goat	vī.ȷ̃ã̄	
028	chien	dog	bí.sì	
029	chose	thing	ní.ŋgà	
030	cinq	five	mìí	
031	cœur	heart	ŋgā.lā	
032	colline	hill	kó.ŋgō	
033	compter	to count	tá.zá	
034	connaître	to know	gè.ɽì	

Table 26: 204-item wordlist (035–065)

	French	English	Phonetic form	Notes
035	corde	cord	kū.là	
036	corne	horn	gā.zù	
037	cou	neck	mí.ndì	
038	couper (couteau)	to cut (knife)	k-ū.jà	
039	couper (hache)	to chop (axe)	k-ū.gu̯à	
040	cracher	to spit	tī.ɓī	
041	creuser	to dig	k-ū.ɗì	
042	cuire	to cook	ndī.ɗī	
043	cuisse	thigh	māā	
044	cul; fondement	bottom	nzà kú.zì	‘foot’ + ‘house’
045	cultiver	to cultivate	nzà.kà	
046	danser	to dance	k-ū.zì	
047	debout (être)	to stand	ɗá.rə́ tà.rú ~ ɗár tà.rú	‘stay’ + ‘upright’
048	dent	tooth	ŋgā.ŋgā	incisors
			gō.gò	premolars, molars
049	deux	two	zí.jó	
050	dire	to say	nzí.lē	
051	donner	to give	k-à.dà	
052	dormir	to sleep	tɔ̄.ɗɔ̀	
053	droite (à)	right	kā vú.lā	
054	eau	water	mà.nɛ̀	
055	écorce	bark	mà.mù	
056	éléphant	elephant	kì.zì	
057	enfant	child	ŋgō.nō	
058	enfanter	to give birth	k-ō.jò	
059	enfler	to swell	k-ūù	
060	entendre	to hear	k-ɔ̀	
061	envoyer	to send	k-ū.là	
062	épine	thorn	kó.nō, mɔ́.ɗɔ̀	
063	étoile	star	kí.ndà.gú	
064	étranger	stranger	ŋgbā	
065	excréments	excrement	kí.zī	

Table 27: 204-item wordlist (066–093)

	French	English	Phonetic form	Notes
066	façonner (pot)	to make (clay pot)	k-ù.ɓà	
067	fagot	firewood	kī.rī	
068	faim	hunger	ɓō	
069	farine	flour	ndù.zú	
070	femme	female	nī.jé	
071	fer	iron	tē.ɱvē	
072	feu	fire	fú.ɗū	
073	feuille	leaf	ká.ɱvà	
074	filet	net	bá.ndà	
075	finir	to finish	tá	
076	flèche	arrow	kù.rà	
077	foie	liver	wū.rù	
078	forger	to forge	k-ō.kù	
079	frapper; battre	to hit	k-ū.ndà	
080	froid (être)	to be cold	kū.lī ù.ku̯à dè	
081	fructifier	to bear fruit	k-à.nù	bear fruit (trees, grain…)
			vā.kà	become numerous
082	fumée	smoke	sāà	
			sì.lí	
083	gauche (à)	left	kā gí.lī	
084	genou	knee	kí.ɽí.kî	
085	graisse	fat	kē.ɱvì	
	huile	oil	kù.ɓù	
086	grand	big	ɓō.ɽì	
087	gratter (se)	to scrape	k-ū.gù	
088	griller	to grill	ɽū.gū	
	faire frire	to fry	nzī.kò	
089	guerre	war	ɗɔ̄ɔ̄	
090	herbe	grass	ŋgì.lì	
091	homme	male	dà vú.là	
092	hyène	hyena	ɓɔ̀.ŋgɔ̀	
093	il	he/she	rō	

Table 28: 204-item wordlist (094–125)

	French	English	Phonetic form	Notes
094	ils	they	rō.gē	
095	intestins	intestines	tì.kpì	
096	je	I (SBJ)	mə̄	
	me	me (OBJ)	mā	
097	jumeaux	twins	nzū.ŋgu̯à	
098	lancer; jeter	to throw	k-ì.nà	
099	langue (organe)	tongue	nzò	
	langue (langage)	language	tà.rà ɓē	
100	laver	to wash	nzì.gbò	
101	long	tall	kā.wù	
102	lourd	heavy	wà	
103	lune	moon	nzó.jì	
104	maison	house	kú.zì	
105	manger	to eat	k-ū.sà	
106	marcher	to walk	ndí.gā	
107	mauvais (être)	to be bad	kà.sù	
108	mère	mother	kóò	
109	montagne	mountain	kó.ŋgō	
110	mordre	to bite	zɔ̄.j̃ɔ̀	
111	mortier	mortar	vì.ɗì	
112	mouche	fly	jú.lū	
113	mourir	to die	k-u̯à	
114	ne...pas	not	gù.nú	
115	neuf; nouveau	new	jã́	
116	nez	nose	kó.nū	
117	noir	to be black	k-ī.lì	
118	nom	name	ɽ̃ī	
119	nombreux (être)	to be many	vā.kà	
120	nombril	navel	kū.mū	
121	nous (INCL)	we	zī.gī	
122	nuit	night	ndū.rí	
123	œil/visage	eye/face	kā.mù	
124	œuf	egg	kā.ɓè	
125	oiseau	bird	jì.lì	

Table 29: 204-item wordlist (126–156)

	French	English	Phonetic form	Notes
126	oncle (maternel)	uncle	nā.nā ~ nā.nū	
127	ongle/griffe	nail	ɽí.kí.zī	
128	oreille	ear	ɱvī	
129	os	bone	jī.ŋgō	
130	panthère	panther	ŋgé.ŋgə̄ sá.kū	'lion' + 'little'
131	peau	skin/fur	ndà.rā	
132	père	father	vū.ɽà	
133	personne	person	dè	
134	petit	small	sá.kū	
135	peur	fear	ɓè.lè	
136	pied	foot	nzà	
137	pierre	rock	kɔ̄.sɔ̄	
138	plaie	wound	dī.jò	
139	planter (arbre)	to plant (tree)	k-ū.dù	
	semer	to sow (seed)	k-ù.sì	
140	plein	full	ɗɔ̄.sɔ̀	
141	pleurer (pleurs)	to cry	k-ōs-ká	'pierce' + 'action of crying'
	larme/pleurs	tears	nɔ̃̄ɔ̃̄	
142	pluie	rain	nzī	
143	poil/plume	hair/feathers	vè.lé	
144	poisson	fish	kā.nzē	
145	pou	louse	ŋgí.sā	
146	poule	chicken	kū.nzá	
147	pourrir	to rot	nzū.mù	
148	prendre	to take	k-ū.nù	
149	profond (être)	to be deep	k-ū.lì	
150	puiser	to draw (water)	k-ō.dò	
151	quatre	four	sɔ̄ɔ́	
152	queue	tail	kī.lá	
153	qui ?	who	nȁ	
154	quoi ?	what	ɗī	
155	racine	root	ŋgī.rà	
156	respirer	to breathe	k-à.vò	

Table 30: 204-item wordlist (157–186)

	French	English	Phonetic form	Notes
157	rester	to stay	k-ī.nzì	
	s'asseoir	to sit down	k-ī.nzì ŋgá.ɽā	'stay' + 'ground'
158	rire	to laugh	tí.gbō	
159	rosée	dew	tā.lù	
160	rouge	to be red	k-ā.kɛ̀	
161	sable	sand	wâ	
162	sagaie	spear	ŋgbā.ɽā	
163	saison des pluies	rainy season	ɓā.rà	
164	saison seche	dry season	ɓā.lā	
165	salive	saliva	wɔ̄.rɔ̄	
166	sang	blood	má.sù	
167	sec (être)	to be dry	nzū.tù	
168	sein	breast	ɱvà	
169	sel	salt	kà.tà	
170	semence/graine	seed/grain	kī.jō	
171	sentir (INTR)	to smell	k-ā.tù	
172	serpent	snake	ní.ŋgá ku̯â	'thing' + 'die'
173	soleil	sun	kā.zà	
174	souffler	to blow	tōō j̃ɛ́.lɛ̀	'blow' (V/186) + 'wind'
175	sucer	to suck	tá.nū	
176	sucré (être)	to be sweet	ndī.tɛ̄	
177	termite	termite	ŋgɔ̄.ɽɔ̀	
178	terre; sol	earth; soil	ŋgá.ɽā	
179	tête	head	zò	
180	téter	nurse	k-ū.lì ɱvà	'suckle' (V/095) + 'breast'
181	tomber	fall	k-ì.sò	
182	tortue	turtle	jí.kò	
183	tous; tout	all	dū.tū	
184	tousser	to cough	ké.sī	
185	tranchant (être)	to be sharp	k-ā.tù	
186	transpercer	to pierce; to stab	k-ō.sò	

Table 31: 204-item wordlist (187–204)

	French	English	Phonetic form	Notes
187	travail	work	kū.là	
188	trois	three	mù.tá	
189	trou	hole	gɔ̄.tɔ̀	
190	tu	you (2SG)	jī	
191	tuer	to kill	tā.lā	
192	un	one	ɗɔ́.jí	
193	urine	urine	kí.zī kà.rɛ̄	‘urine’ + ?
194	venir	to come	k-ī.ɗè	
195	vent	wind	j̃ɛ̃́.lɛ̀	
196	ventre	stomach	kē.mè	
197	viande/chair	meat/flesh	zã̄, tá.ɗù	
198	village	village	ɓē	
199	voir	to see	k-ā.kù	
200	voler (oiseau)	to fly	ɓĩ̀	
201	voler; dérober	to steal	ɓō.gò	
202	vomir	to vomit	tú.ɗē	
203	vouloir	to want	ndī.gì	
204	vous	you (2PL)	sè.gè	